

Lecture 03: Randomized controlled trials I

PPHA 34600
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From last time: selection is an issue

Recall that there are lots of things we want to estimate.

We need to get around selection bias to do this.

In other words, we need:

$$E[Y_i(1)] = E[Y_i(1)|D_i = 1] = E[Y_i(1)|D_i = 0]$$

and

$$E[Y_i(0)] = E[Y_i(0)|D_i = 0] = E[Y_i(0)|D_i = 1]$$

Regression equivalent:

$$E[\varepsilon_i|D_i] = 0$$

Random assignment as a solution

When treatment status is randomly assigned,

$$F(X, \varepsilon | D = 1) = F(X, \varepsilon | D = 0) = F(X, \varepsilon)$$

In words:

The distribution of **both** observables (X s) **and** unobservables (ε s) is the same for treated and untreated units!

There is **no selection problem** by construction!

Again, but mathier

When D , treatment, is **randomly assigned**:

- D is independent of $Y(0)$ and $Y(1)$
- The distribution of $Y_i(0)|D_i$ is equal to the unconditional distribution
- The distribution of $Y_i(1)|D_i$ is equal to the unconditional distribution
- $E[Y_i(1)|D_i = 1] = E[Y_i(1)]$
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As a result:

$$\begin{aligned}\tau^{ATE} &= E[Y_i(1)] - E[Y_i(0)] \\ &= E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] \\ &= E[Y_i|D_i = 1] - E[Y_i|D_i = 0]\end{aligned}$$

Under randomization:

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We can easily estimate this from data:

$$\hat{\tau}^{ATE} = \overline{Y(1)} - \overline{Y(0)}$$

We can estimate the ATE simply from the difference in means between treated and “control” group.

This bears repeating

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Obvious (?) caveat: We still can't get τ_i , because we only observe i once.

Evaluating an RCT

This is not a class on how to do RCTs

- As always, the devil is in the details
- Field experiments are *hard!*
- But supposing you've got one...

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Basic RCT checklist

- ☐ Verify random assignment
- ☐ Check compliance with treatment
- ☐ Estimate the ATE (or other things...)
- ☐ Examine heterogeneity in impacts

Application: Microfinance in India

Policy challenge:

- Microfinance has been posited as a way out of poverty...
- ...but there's also been some evidence that it doesn't work super well

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Program:

- Implementer: Spandana, a large indian MFI
- Provides group loans to women

What is this experiment trying to learn?

When running an RCT, you want to have a “research question” in mind:

What is the causal effect of microfinance on businesses?

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When running an RCT, you want to have a “research question” in mind:

What is the causal effect of microfinance on businesses?

Why do we need an RCT to study this?

- MFIs target some locations over others
- People who select into microfinance look different from those who don't
- Others?

Microfinance: The experiment

The MFI implemented an RCT!

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This is a **randomized rollout** design:

- Enrollment couldn't all be done at once
- The MFI phased in enrollment across Hyderabad
- The unit of randomization is the **neighborhood**
 - Why?

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- The unit of randomization is the **neighborhood**
 - Why?
- 52 treatment neighborhoods, 52 control neighborhoods
- Control units were treated 3-4 years later

Outcomes of interest

Outcome data measured at baseline, 15-18 month midline, 3-4 year endline

Outcomes of interest:

- Treatment status
- Credit amounts (from various sources)
- Business operations: investments; profits, self-employment
- Income; work time; consumption
- Social effects

Note that randomization happened at the **area** level, but data are collected at the **individual** level...

... more on this later!

Verifying random assignment

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Testing whether randomization was effective:

- We want T and C to be similar on observables **and** unobservables
- We can only test this for observables
- To check this, we “test for balance”:
- Compare mean outcomes for T vs. C *at baseline* (before treatment) or in fixed characteristics
 - Implementation: Regress $Y_i^{baseline} = \alpha + \tau D_i + \nu_i$

Balance in microfinance?

TABLE 1A—BASELINE SUMMARY STATISTICS

	Control group			Treatment – control	
	Obs. (1)	Mean (2)	SD (3)	Coeff. (4)	p-value (5)
<i>Household composition</i>					
Number members	1,220	5.038	(1.666)	0.095	0.303
Number adults (≥ 16 years old)	1,220	3.439	(1.466)	−0.011	0.873
Number children (< 16 years old)	1,220	1.599	(1.228)	0.104	0.098
Male head	1,216	0.907	(0.290)	−0.012	0.381
Head's age	1,216	41.150	(10.839)	−0.243	0.676
Head with no education	1,216	0.370	(0.483)	−0.008	0.787
<i>Access to credit</i>					
Loan from Spandana	1,213	0.000	(0.000)	0.007	0.195
Loan from other MFI	1,213	0.011	(0.103)	0.007	0.453
Loan from a bank	1,213	0.036	(0.187)	0.001	0.859
Informal loan	1,213	0.632	(0.482)	0.002	0.958
Any type of loan	1,213	0.680	(0.467)	0.002	0.942

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Three things to check for:

- 1 Did they test for all outcome variables?
- 2 Are differences statistically significant?
- 3 Are magnitudes economically meaningful?

Did assignment to treatment affect treatment status?

Trying to verify whether...

- Units assigned to treatment were actually treated
- Units assigned to control were *not* treated

There is often substantial non-compliance. We'll talk more about exactly how to deal with this issue next time.

Was there compliance in the MFI experiment?

TABLE 2—CREDIT

	Spandana (1)
<i>Panel A. Endline 1</i>	
<i>Credit access</i>	
Treated area	0.127*** (0.020)
Observations	6,811
Control mean	0.051
Hochberg-corrected <i>p</i> -value	

Sort of:

- 5.1 percent of control households had credit from Spandana
 - 12.7 percent of treatment households had credit from Spandana
- The experiment **affected** treatment, but not perfectly!

Thinking about non-compliance

We will treat this more formally next time

For now, non-compliance changes the interpretation of our estimates:

Rather than asking "What does a loan do to business activities?" ...
... we're asking "What does adding a MFI branch do to business activities?"

This may be the policy-relevant quantity

We want to estimate the ATE

Recall that the ATE is just:

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Regression is a convenient way to do this:

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

Since our $E[\varepsilon|D_i] = 0$ assumption is satisfied (why?), $\hat{\tau} = \hat{\tau}^{ATE}$

Estimating the impacts of microfinance

The actual equation in the paper is a little more complicated:

$$y_{ia} = \alpha + \tau Treat_{ia} + \gamma \mathbf{X}_a^{\text{baseline}} + \varepsilon_{ia}$$

where:

- y_{ia} are outcomes for household i in area a
- α is a constant
- $Treat_{ia}$ is a treatment dummy (think D_i)
- $\mathbf{X}_a^{\text{baseline}}$ is a set of baseline area controls
- ε_{ia} is an error term

What is this equation estimating?

$$y_{ia} = \alpha + \tau \textit{Treat}_{ia} + \gamma \mathbf{X}_a^{\text{baseline}} + \varepsilon_{ia}$$

This differs from our basic regression a bit:

- There's an i and an a
- We have $\gamma \mathbf{X}_a^{\text{baseline}}$
- We're not (exactly) getting the ATE because of non-compliance

Let's unpack each of these in turn...

Randomization by area, data on individuals

We have i -ndividual level data, but a -rea level randomization

Randomizing at a higher level of aggregation is common:

- Some questions can't be answered at i level (no personal bank branches)
- Ethics concerns: can sometimes delay implementation for a whole group; hard for individuals
- Reduce spillovers (more on this later)

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Randomizing at a higher level affects the analysis:

- Interpretation is different (what exactly is treatment?)
- Getting standard errors right requires either:
 - 1 Estimate i -level effects, but cluster at a -level
 - or
 - 2 Averaging outcomes at the group level (weight by individuals per group)

Adding controls

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→ Why?

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- Controlling for X_i should not affect $\hat{\tau}$
→ Why?
- Controlling for X_i will affect the standard error on $\hat{\tau}$
→ Why?

do **not** control for post-treatment outcomes

First rule of RCT club:

- Do **not** control for post-treatment outcomes
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- If treatment affects these outcomes, you can get bias!

Simple example:

- Suppose microfinance impacts business ownership
- By random assignment, households with and without loans have the same potential income
- Once we condition on business ownership, this is no longer true!

We can use simulated data to think about this

Type of household	Potential business ownership		Potential income		Average earnings by ownership	
	Without MF	With MF	Without MF	With MF	Without MF	With MF
Never owner	No	No	1,000	1,500		
Moved by MF	No	Yes	2,000	2,500		
Always owner	Yes	Yes	3,000	3,500		

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- The return to MFI is 500 for everyone...
 - But once we condition on ownership, it looks like the return is 0!
- This is because we don't have random assignment **within** ownership!

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Do not control for post-treatment outcomes!

What are we actually estimating?

For now, non-compliance changes the interpretation of our estimates:

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One more definition:

- We're estimating an **intent to treat (ITT) effect** for Q1
- But estimating an **ATE** for Q2!

What do we find?

TABLE 3—SELF-EMPLOYMENT ACTIVITIES: REVENUES, ASSETS, AND PROFITS (*All households*)

	Assets (stock) (1)	Investment in last 12 months (2)	Expenses (3)	Profit (4)	Has a self- employment activity (5)	Number of self- employment activities (6)	Has started a business in the last 12 months (7)	Has closed a business in the last 12 months (8)	Index of dependent variables (9)
<i>Panel A. Endline 1</i>									
Treated area	598 (384)	391* (213)	255 (1,056)	354 (314)	0.0083 (0.0215)	0.018 (0.0380)	0.009 (0.006)	0.002 (0.008)	0.0357 (0.0188)
Observations	6,800	6,800	6,685	6,239	6,810	6,810	6,757	2,352	6,810
Control mean	2,498	280	4,055	745	0.349	0.503	0.047	0.037	0.000
Hochberg-corrected <i>p</i> -value									0.175

What do we find?

TABLE 4—INCOME

	Self employment (profit) (1)	Daily labor/salaried (2)	Index of dependent variables (3)
<i>Panel A. Endline 1</i>			
Treated area	354 (314)	−526 (358)	−0.0501 (0.0459)
Observations	6,239	6,827	6,832
Control mean	745	2,988	0.000
Hochberg-corrected <i>p</i> -value			>0.999

We can also estimate heterogeneous effects

Heterogeneous effects are straightforward:

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We typically estimate these in two ways:

- 1 Add an **interaction term** to the regression:

$$y_i = \alpha + \tau Treat_i + \gamma Treat_i \cdot X_i + \delta X_i + \varepsilon_i$$

→ Make sure to add both the interaction and the base term

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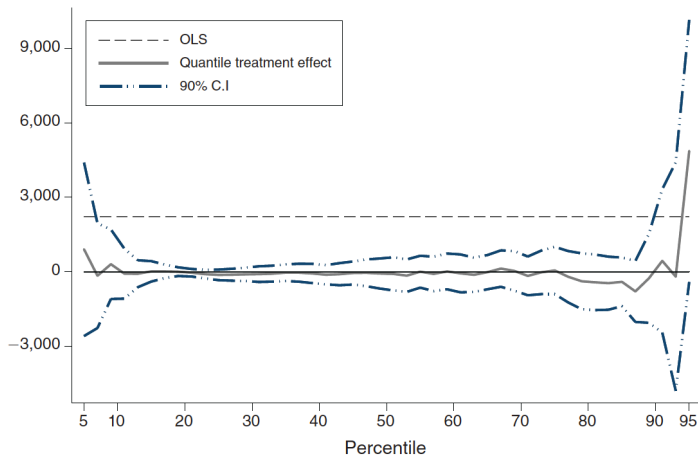
→ Make sure to add both the interaction and the base term

- 2 Estimate the regression **separately** by heterogeneity

→ Equivalent to a *fully* interacted model

Estimate heterogeneity by pre-determined characteristics only!

What do we find?



A note on assumptions for the RCT

We still need several assumptions for the RCT to work:

- $E[Y_i|D_i = 1] = E[Y_i|D_i = 0]$
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A note on assumptions for the RCT

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 - We “get this” via randomization, but only in expectation
- Perfect compliance
 - Kinda. More on this next class

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 - We “get this” via randomization, but only in expectation
- Perfect compliance
 - Kinda. More on this next class
- No spillovers: “SUTVA”
 - Stable Unit Treatment Value Assumption: D_i doesn't affect j 's potential outcomes
 - Kinda. More on this in two classes

TL;DR:

- ① RCTs are great!
- ② Experiments solve our selection problem
- ③ Be very careful with adding controls

For next class

Topics:

- Randomization II: Noncompliance

Reading: Fowlie, Wolfram, et al (2018). Carefully read:

- 1: Introduction
- 3: Empirical setting and experimental design
- 4: Data and methodology
- 5: Main results
- 8: Conclusion
- Appendix C